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Citation: *The Physics Teacher* **41**, 158 (2003); doi: 10.1119/1.1557504

View online: <http://dx.doi.org/10.1119/1.1557504>

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The Gauss Rifle and Magnetic Energy

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With the advent of cheap and easy-to-come-by NdFeB magnets, it has become possible to design a number of simple but effective demonstrations of magnetic force. One such demonstration, dubbed the “gauss rifle,” is a type of linear magnetic accelerator. It is relatively easy to assemble and involves a rapid and dramatic increase in kinetic energy of the steel ball bearings used in the demonstration. This makes the demonstration a good attention getter, setting the stage for a discussion of a number of physics topics, including conservation of energy, magnetic energy, and magnetic force. It also has the potential for becoming a laboratory experiment since the materials are relatively cheap, there is some challenge in the arrangement of the magnets, and the performance of the accelerator can be characterized by measuring the initial and final velocities of the bearings. Finally, by using freely available finite element magnetics software, it is even possible to make predictions of the final velocity for different configurations of the magnets.

Apparatus

The gauss rifle is best constructed using a wooden ruler with a slot down the center, four 12-mm square gold-plated NdFeB magnets, plenty of cellophane (or electrical) tape, and nine 0.795-cm radius nickel-coated steel ball bearings. It is best to buy a ruler with either no metal strip or with a brass one to avoid problems with placing the magnets on the ruler. Plastic rulers are also possible, although if cellophane tape is used, the tape sticks better to the wood. The NdFeB

magnets of this dimension are all of similar quality. The strength of the magnetization in the material per unit volume is usually given as the energy product in units of MGOe (read mega gauss-oersteds). A typical NdFeB magnet available online has an energy product of about 35 MGOe. This is useful to know when trying to simulate the demonstration using finite element analysis. Similarly, the steel bearings are usually made of a low-carbon steel, such as 1020 steel (containing 0.20% carbon by weight), which has a known, nonlinear magnetic permeability.

The magnets should be placed squarely over the center of the ruler, with their north pole faces all pointing in the same direction, lying at the 2.5-, 5.0-, 7.5-, and 10-in marks, respectively, which leaves approximately 2.0 in between magnets and between each end magnet and the end of the ruler. A magnetic compass or other pole detector is necessary to identify the pole faces on the square magnets. To secure each magnet to the ruler in a relatively permanent fashion, electrical tape can be wrapped around the magnet and ruler at least two times. Once the magnets have been secured, the tape needs to be trimmed off of the faces of the magnet so that the bearings can freely come into contact with the poles.

Once the magnets are secured in place, eight of the steel bearings are placed two by two, touching the north pole faces of each of the four magnets. Failure to secure the magnets tightly before beginning this phase is guaranteed to produce a tangled mess of magnet chips, bearings, and crumpled tape. The ninth bearing is the incoming projectile, which is released



Fig. 1. Gauss rifle, with PASCO photogates for measuring the incoming and outgoing bearing velocities.

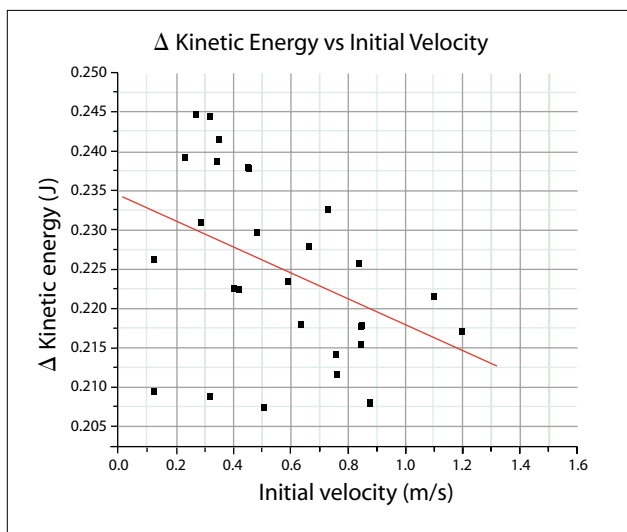


Fig. 2. Observed difference in kinetic energy of the incoming and outgoing bearings as a function of the initial velocity.

with a low velocity toward the south pole of the first magnet. The magnetic force on the incoming bearing causes it to accelerate and strike the magnet with sufficient force that the kinetic energy transferred to the second bearing on the opposite side allows it to break contact and move toward the second magnet. Since this bearing is able to move to a position much closer to the second magnet, it experiences a gain in energy and smashes into the second magnet with even greater force. This process is repeated for each successive outer bearing, and the final result is that the last bearing jumps off of the ruler like a bullet from a rifle, hence the name. The entire rearrangement of the system occurs in roughly 1/20th of a second, and the ball leaves the system with such a large velocity that it has quite an impact, both literally and figuratively.

Velocity Measurements

We can characterize the rifle's performance as an accelerator by measuring the input and output velocities. This is easily done using photogates at the front and rear of the assembly. The bearings in this case had a diameter of 1.59 cm, with a mass of 16.4 g. Using Vernier's LabPro 2.1.1 interface and software, and PASCO photogates, I made a number of measurements of the bearing velocities. The setup is shown in Fig. 1. A few comments are in order before presenting the results. I used electrical tape to firmly secure the magnets to the ruler. The first several "shots" didn't fire. Thereafter, over the first five to 10 shots, the final velocity gradually got larger and larger. Apparently, the magnets were *too* secure in the beginning, and the tape restricting their motion absorbed a significant amount of the kinetic energy during the first impact. Second, I found that the lower the initial velocity, the more efficient the rifle. Indeed, there was a slightly negative correlation between initial and final velocities for initial velocities ranging from 0.2 m/s to 1.2 m/s. It is speculated that the lower initial velocity meant that there was less sideways motion. This effect was

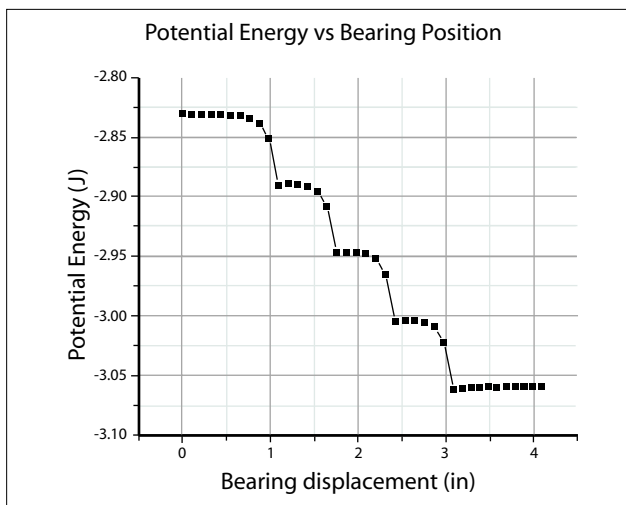


Fig. 3. Change in the potential energy of the rifle as a function of the accumulated displacement of the steel bearings.

more pronounced in the plot of ΔKE versus initial velocity, as seen in Fig. 2. Thus, the rifle as designed is not a good test of the work-energy theorem!

Nevertheless, we do get an approximate measure of the increase in kinetic energy of the moving bearing. From the graph, we find that the average increase in kinetic energy for the 28 data points considered was 0.224 J, with a standard deviation of 0.012 J. For an initial velocity of 0.5 m/s, this corresponds to a final velocity of about 5.3 m/s and an approximately ten-fold increase in the velocity.

Energy Analysis

Assuming a perfect transfer of energy for each collision, we can conclude that the kinetic energy gain experienced by the moving bearings comes at the expense of a decrease in their potential energy as they are knocked from their original position (one bearing removed from the north pole) and into direct contact with the south pole of the neighboring magnet. Since this readjustment occurs four times (counting the incoming bearing), the final bearing has acquired four additional “units” of kinetic energy, where a “unit” is the amount of energy gained by the bearing as it falls from the “excited state” position to the “ground state” position. In this light, it is possible to consider the gauss rifle as a visible demonstration of the energy released during “quantum” state transitions.

A quantitative energy analysis in electromagnetic

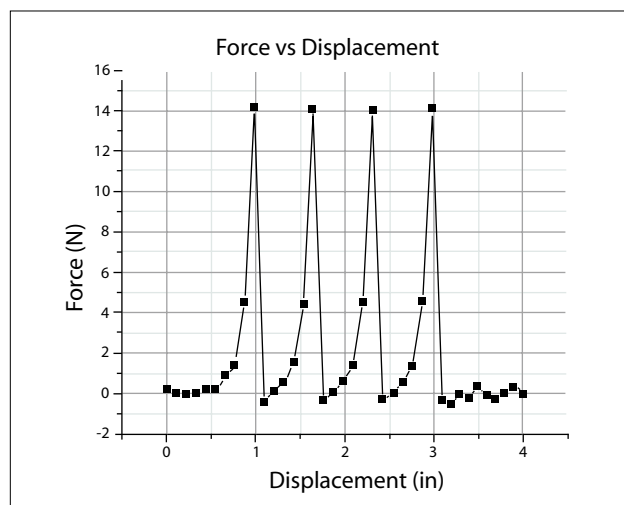


Fig. 4. Force on the moving bearing as a function of the accumulated displacement of the steel bearings.

problems requires knowledge of the field(s) throughout space. Starting from the energy required to induce a current I in an inductor with self-inductance L , equal to

$$W = \frac{1}{2} LI^2, \quad (1)$$

it is possible to develop an expression in terms of the magnetic field, B , for the total energy in a volume filled with current sources only¹

$$W = \frac{1}{2\mu_0} \int_v B^2 dx^3. \quad (2)$$

When permanent magnets are used instead of current sources and nonlinear magnetic materials are also present, this must be rewritten in terms of both B and the applied field, H , giving what is known as the coenergy,^{2,3}

$$W_c = \int_v \left[\int_0^H B(H) \cdot dH \right] dx^3. \quad (3)$$

The inner integral is necessary when nonlinear magnetic materials are present in order to account for the energy required to build the fields up from nothing. The coenergy is the appropriate form of the field energy to use for constant current sources, which are a valid model for linear, permanent magnets.^{3,4} It is important to note that for an object moving from A to B, an *increase* in the field coenergy corresponds to

a *decrease* in potential energy of the object, since the energy gained by the field from the increased magnetization of the object is twice that lost by the field as it does mechanical work on the object.

This problem was modeled using the freely available finite element magnetics software FEMM 3.1, assuming an axisymmetric geometry for ease of use. The manufacturer of the magnets could not give a precise energy product for the magnets, but estimated it to be 35 MGOe. The software has 37 MGOe Nd-FeB in its materials library, and this was used in the calculation. The magnets are in fact square instead of cylindrical, and so to keep the volume the same, the height was kept at 1.2 cm, but a cylinder radius of 0.677 cm was used. The nickel-plated steel bearings were made of a low-carbon steel, presumed to be 1020 steel, whose properties are also available in the materials library. The program has a built-in application extender, based on the programming language Lua, that allows the user to calculate the fields for a series of static configurations, calculate the coenergy for each, and write it to a file. The potential energy was obtained by changing the sign of the coenergy. The force as a function of position can be obtained from the *positive* ratio of the change in coenergy over the change in position of the bearing. The results of the calculation are displayed in Figs. 3 and 4. The displacement axis in each figure records the *accumulated* displacement of the bearings as the rifle propels each bearing from left to right. Thus, for example, in Fig. 4, the force is acting on the first bearing over the first inch of displacement, on the second bearing over the next 0.67 in, etc. The step-like behavior in the potential energy is clearly displayed. In addition, the calculated total change in energy of the system, $E = 0.230$ J, agrees with the experimentally determined change in the energy of the bearings to within one standard deviation. The force is seen to increase rapidly as a function of decreasing separation, which is to be expected for forces on induced magnetic moments.

Summary

The gauss rifle as commonly constructed is an excellent demonstration of how kinetic energy can be extracted from a metastable arrangement of magnets and ferromagnetic bearings. In addition to its usefulness in illustrating magnetic force and energy, it serves

as a great tool for illustrating how transitions from excited to ground states can produce a sudden and predictable amount of energy being released from the system. The system can be investigated in a more quantitative way using photogate sensors, and a value for the energy released by the system can be obtained. Unfortunately, the efficiency of energy release is not independent of the initial velocity, with lower initial velocities generally producing greater net releases of kinetic energy. This deficiency may be correctable with further adjustments in the design of the rifle. A very accurate prediction of the amount of energy released when the rifle is fired can be obtained through the use of a freeware finite element magnetics software package, FEMM 3.1. In addition, the program can be used to predict the force experienced by the bearings and how the fields change as the bearings move along the ruler, making the difficult association of energy and fields more “real” for the students. Although the expression used to obtain the coenergy of the field is too advanced for an introductory physics course, the *idea* of the energy being stored in the fields is not, so that such a tool as FEMM 3.1 should not be beyond the students’ ability to implement.

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PACS codes: 01.40D, 01.40Eb, 01.50Fb, 01.50M

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